

Project Name: OpenCart (Frontend)

Test Plan

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1. Introduction

The purpose of this Test Plan is to define the testing approach, scope, objectives, resources, responsibilities, test environment, schedule, risks, and deliverables for testing the OpenCart E-commerce Application. The objective of testing is to verify that the application meets the functional and non-functional requirements specified in the Functional Requirement Specification (FRS). Testing will focus on validating major e-commerce functionalities such as user registration, login, product search, product display, product comparison, wishlist, shopping cart, checkout, and reporting. The testing process will help identify defects at an early stage and ensure that the application is reliable, secure, usable, compatible, and ready for release.

2. Test Item

The primary test item is the **OpenCart E-Commerce Application**, including its customer-facing website and administrative functionalities.

The application will be tested to verify that customers can successfully browse products, manage their accounts, add products to the cart or wishlist, place orders, and complete the checkout process. Administrative functionalities such as product, inventory, customer, order, and report management will also be tested.

2.1 Feature to be Tested

The following features will be included in the testing scope:

1. **User Registration**
    - New customer registration
    - Mandatory field validation
    - Valid and invalid input validation
    - Duplicate email validation
2. **Login Functionality**
    - Valid login
    - Invalid username/email and password
    - Empty field validation
    - Login/logout behavior

3. **Forgot Password**
    - Password recovery request
    - Email validation
    - Invalid email handling
    - Password reset functionality
4. **Product Search**
    - Search using valid keywords
    - Search using invalid keywords
    - Partial keyword search
    - Search result accuracy
5. **Product Display**
    - Product name
    - Product image
    - Price
    - Description
    - Availability/stock status
    - Product details
6. **Product Comparison**
    - Add products for comparison
    - Compare product specifications
    - Remove products from comparison
7. **Shopping Cart**
    - Add product to cart
    - Update quantity
    - Remove product
    - Price calculation
    - Cart total calculation

8. **Wishlist**
    - Add product to wishlist
    - Remove product
    - Wishlist visibility for logged-in users
9. **Checkout**
    - Customer information
    - Billing address
    - Shipping address
    - Shipping method
    - Order summary
    - Order confirmation
10. **Payment**
    - Available payment methods
    - Successful payment
    - Failed payment
    - Payment validation

11.Report Management

- Order reports
- Sales reports
- Product reports
- Customer-related reports

12.Logout

- Successful logout
- Session termination
- Access restriction after logout

13.UI/Usability

- Page layout
- Navigation
- Buttons
- Links
- Error messages
- Form alignment

2.2 Feature Not to be Tested

The following items are outside the scope of the current testing cycle unless specifically required:

- Internal implementation of third-party payment gateways
- Internal functionality of external shipping providers
- Internal functionality of external email/SMS services
- Third-party infrastructure that is not controlled by the project team
- Source-code unit testing performed by developers
- Production server infrastructure testing
- Features that are not included in the approved FRS
- Future features that are planned for subsequent release

3. Quality Objectives

The main quality objective is to ensure that the OpenCart application meets the approved business and functional requirements and provides a reliable experience to customers and administrators.

The testing objectives are to:

- Verify that all specified requirements are implemented correctly.
- Identify functional and non-functional defects.
- Verify that critical business workflows work correctly.
- Ensure that customer and administrator functionalities operate as expected.
- Verify proper validation and error handling.
- Ensure accurate product, inventory, price, and order information.
- Verify the security of authentication and authorization.
- Verify browser and device compatibility.
- Verify application usability and navigation.
- Verify that critical and high-severity defects are resolved before release.
- Minimize the risk of production failures.

4. Test Methodologies

Testing will be performed using a combination of manual and, where applicable, automated testing techniques. Testing will be based on the requirements documented in the FRS and will include positive, negative, functional, regression, usability, compatibility, and other appropriate testing techniques.

4.1 Testing Level

The following testing levels will be considered:

1. Unit Testing:
Testing of individual components or modules. This activity is primarily performed by developers.

2. Integration Testing:
Testing interactions between different modules, such as product, cart, checkout, payment, order, and inventory modules.

3. System Testing:
Testing the complete OpenCart application against the specified requirements.

4. User Acceptance Testing (UAT):
Testing performed to confirm that the system meets business requirements and is acceptable for release.

4.2 Test Strategy

The testing strategy will include the following approaches:

- **Functional Testing** – Verify that each feature works according to requirements.
- **Smoke Testing** – Verify that the major application functions are stable enough for detailed testing.
- **Sanity Testing** – Verify specific functionality after changes or fixes.
- **Regression Testing** – Verify that existing functionality is not affected by changes.
- **Retesting** – Verify that reported defects have been successfully fixed.
- **Positive Testing** – Verify expected behavior using valid inputs.
- **Negative Testing** – Verify system behavior using invalid or unexpected inputs.
- **Boundary Value Analysis** – Test values at and around defined boundaries.
- **Equivalence Partitioning** – Divide input data into valid and invalid classes.
- **Exploratory Testing** – Explore the application to identify unexpected defects.
- **Usability Testing** – Verify ease of navigation and user interaction.
- **Compatibility Testing** – Verify application behavior across supported browsers and devices.
- **Security Testing** – Verify authentication, authorization, session handling, and basic security controls.
- **Performance Testing** – Verify application behavior under expected load where applicable.

5. Roles and Responsibilities

Role	Responsibilities
Project Manager	Project planning, coordination, schedule management, and overall project monitoring.
Business Analyst	Requirement gathering, requirement clarification, and FRS maintenance.
QA/Test Lead	Test planning, test strategy, task allocation, review, and test reporting.
QA/Test Engineer	Test case preparation, test execution, defect reporting, retesting, and regression testing.
Developer	Code development, defect fixing, unit testing, and technical support to QA.
Database Administrator	Database setup, data management, backup, and database-related support.
System Administrator/DevOps	Test environment setup, deployment, configuration, and environment support.
Business/User Representative	Requirement validation and User Acceptance Testing.

6. Entry and Exit Criteria

Entry and exit criteria define the conditions required to start and complete a testing activity.

6.1 Entry Criteria

Testing can begin when:

- Approved FRS is available.
- Testable build/application is available.
- Required functionalities are implemented.
- Test environment is configured and accessible.
- Required test data is available.
- Test cases have been prepared and reviewed.
- Required testing tools are available.
- Dependencies required for testing are available.
- Major development blockers have been resolved.
- QA team has received the required application access and credentials.

6.2 Exit Criteria

Testing can be considered complete when:

- All planned test cases have been executed.
- Critical and high-severity defects are resolved or formally accepted.
- Required regression testing is completed.
- Retesting of fixed defects is completed.
- No open blocker/critical defects remain without approval.
- Test execution results are documented.
- Requirement coverage is achieved as per project expectations.
- Test summary report is prepared.
- Stakeholders approve the testing results.
- Application is considered suitable for release.

7. Suspension Criteria and Resumption Requirement

Testing may be suspended when a critical application failure prevents further testing, the test environment becomes unavailable, the application build is unstable, major functionality is missing, required test data is unavailable, or critical external dependencies are unavailable. Testing may also be suspended if a large number of blocking defects prevent meaningful execution of planned test cases.

Testing will resume after the identified blocking issue has been resolved, a stable application build has been provided, the test environment is restored, required test data or external dependencies are available, and the QA team has confirmed that testing can continue. After resumption, appropriate smoke or sanity testing should be performed before continuing with the remaining test activities.

8. Resource and Environment Needs

8.1 Resources

The following resources may be required for testing:

Hardware

- Desktop/Laptop computers
- Minimum required RAM and processor according to application requirements
- Stable internet connection
- Mobile devices for responsive testing where required

Software

- Supported operating system
- Web browsers
- OpenCart application
- Database server
- Web server
- Test management tools
- Defect management tools

Human Resources

- QA Engineers/Testers
- QA/Test Lead
- Developers
- Business Analyst
- Project Manager
- System/Database Administrator

8.2 Testing Tools

The following tools may be used depending on project requirements:

Tool	Purpose
Process	Tool
Test Case creation	Microsoft Excel
Test Case Tracking	Microsoft Excel
Test Case Execution	Manual
Test Case Management	Microsoft Excel
Defect Management	Jira
Test Reporting	Microsoft Excel & Jira

8.3 Configuration Management

Configuration management will be used to control application builds, test cases, test data, test scripts, environment configurations, and related documents. Approved versions of test cases and test documents should be maintained to ensure traceability. Any changes to requirements, application configuration, test cases, or test scripts should be properly documented and version controlled. Source code and automation scripts, where applicable, should be maintained using a version control system such as Git.

8.4 Test Environment

Testing will be performed in a dedicated test/staging environment that closely represents the expected production environment. The environment should contain the required OpenCart application, web server, database, supported browsers, network configuration, test accounts, and test data. The environment should be stable and accessible to the QA team throughout the testing cycle.

Example environment:

Component	Example
Application	OpenCart E-Commerce Application
Operating System	Windows/Linux
Browser	Google Chrome, Firefox, Microsoft Edge
Web Server	Apache
Database	MySQL
Testing Type	Manual
Defect Tool	Jira
Version Control	Git/GitHub

9. Deliverables

The following testing deliverables will be produced during the testing lifecycle:

1. Test Plan
2. Test Scenarios
3. Test Cases
4. Test Data
5. Requirement Traceability Matrix (RTM)
6. Test Execution Report
7. Defect/Bug Reports
8. Defect Tracking Report
9. Retesting Results
10. Regression Testing Results
11. Screenshots/Evidence of Testing
12. Automation Test Scripts, where applicable
13. Test Summary Report
14. UAT Support Documents
15. Final Test Closure Report

10. Risks & Assumptions

Risks

Potential risks include unclear or frequently changing requirements, unstable test environments, delays in receiving application builds, dependency on third-party payment or shipping services, database or server failures, insufficient test data, limited testing time, limited testing resources, high numbers of defects, browser compatibility issues, and unexpected integration failures. These risks may affect test coverage, schedule, and release decisions.

Assumptions

It is assumed that the approved FRS will be available before test planning and test case preparation. It is assumed that the application build provided to QA will contain the required functionalities and will be reasonably stable for testing. Required test environments, user

accounts, test data, browsers, tools, and system access will be provided on time. It is also assumed that developers and other stakeholders will be available to clarify requirements and resolve reported defects. Any changes to approved requirements will be communicated to the QA team and evaluated for their impact on testing.

11. Approvals

The Test Plan should be reviewed and approved by the appropriate project stakeholders before formal testing begins. Approval indicates that the stakeholders have reviewed the planned testing scope, objectives, methodology, resources, schedule, risks, and acceptance criteria.

Name	Role	Responsibility	Signature	Date
	Project Manager	Project Approval		
	QA/Test Lead	Test Plan Approval		
	Business Analyst	Requirement Approval		
	Development Lead	Technical Approval		
	Business/User Representative	UAT Approval		